Predicting US Domestic Flight Delays: A Comparative Study of Classical ML and Neural Networks with Model Compression

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Abstract

Flight delay prediction is a high-impact problem in aviation operations, affecting millions of passengers and costing airlines billions annually. This paper presents a machine learning pipeline for predicting US domestic flight delays using the 2023 US Civil Flights dataset combined with daily meteorological data from 364 airports. We implement and compare three models — Random Forest, Support Vector Regression (SVR), and a Multi-Layer Perceptron (MLP) — across both binary classification (delayed >15 min) and regression (exact delay in minutes) tasks. We additionally investigate coreset selection as a data reduction strategy, demonstrating that a 50% random sample achieves AUC within 0.002 of full-data training. Our best models achieve AUC $\approx$ 0.95 and $R^2 \approx 0.95$, improving on published benchmarks that do not incorporate weather features.

1 Introduction

Flight delays impose significant costs on passengers, airlines, and airports. According to the FAA, a delay is officially reportable when a flight arrives more than 15 minutes behind its scheduled time. In 2023, approximately 21% of US domestic flights were delayed, resulting in over $30 billion in direct costs to airlines and passengers.

Accurately predicting whether a flight will be delayed — and by how many minutes — has applications in crew scheduling, gate management, passenger notification, and fuel planning. This project addresses the problem as both a classification task (delayed or not) and a regression task (minutes late), using flight schedule features combined with weather observations at the departure airport.

The primary contributions of this work are:

- A reproducible ML pipeline on 6.7 million real 2023 flight records combined with NOAA weather data
- A direct comparison of Random Forest, SVR, and MLP on both classification and regression
- An empirical coreset selection study showing how much data is actually needed for reliable predictions

2 Related Work

Cai et al. (2022) [1] model flight delays as a propagation problem over a time-evolving airport graph. Each airport is a node and each flight is a directed edge; delays cascade when the same aircraft is reused across routes. Their graph neural network approach achieves strong prediction accuracy on US domestic data. This work motivates our use of departure delay as the primary input feature, as it directly captures this cascading effect.

AlBassam & AlShahrani (2025) [2] systematically benchmark six classical ML classifiers on flight delay prediction with a focus on class imbalance. They find Random Forest with balanced resampling achieves AUC = 0.87 and accuracy = 0.90. Their setup — single-source flight data without weather features — serves as a direct baseline for our results.

Grinsztajn et al. (2022) [3] benchmark tree-based models against deep learning architectures across 45 heterogeneous tabular datasets at NeurIPS 2022. They demonstrate that Random Forest and XGBoost consistently match or outperform neural networks on tabular data due to the irregular decision boundaries and heterogeneous feature scales that characterise such datasets. This finding directly predicts and explains why our RF and MLP achieve near-identical AUC.

3 Dataset

3.1 Primary Dataset — 2023 US Civil Flights

We use the *2023 US Civil Flights, Delays, Meteo and Aircraft* dataset [4], publicly available on Kaggle under an open government licence. The main file `US_flights_2023.csv` contains approximately 6.7 million rows and 24 columns, covering every domestic US flight operated in 2023. Key columns include airline carrier, departure airport, scheduled and actual departure/arrival times, flight duration, and delay in minutes.

3.2 Weather Dataset — NOAA Meteorological Data

The file `weather_meteo_by_airport.csv` contains 132,860 rows of daily weather observations for 364 US airports. Columns include mean temperature, precipitation, wind speed, snow depth, and atmospheric pressure. This dataset is structurally and semantically separate from the flight data — it originates from meteorological measurement systems rather than flight operations systems. The two datasets are joined on the composite key `(Dep_Airport, FlightDate)`, attaching departure-day weather conditions to each flight record.

3.3 Exploratory Data Analysis

After removing cancelled flights and rows with missing delay values, the working dataset contains approximately 480,000 records (using a 500k-row sample for D1; full data used for D2+). Key findings from EDA:

- **Class imbalance:** 21.5% of flights are delayed (>15 min), creating a 78.5/21.5 split
- **Schedule rot:** Delay rate rises from ≈5% for morning departures to ≈35% for evening departures as delays cascade across the day
- **Airline variation:** Delay rates range from ≈10% (Hawaiian Airlines) to ≈30% (budget carriers)
- **Strong linear predictor:** Departure delay and arrival delay have Pearson $r \approx 0.97$
- **Weather signal:** Wind speed and temperature show moderate correlation with delay rate

4 Methodology

4.1 Feature Engineering

We construct 12 input features — 7 from the flight dataset and 5 from the weather dataset:

Target variables: Binary classification target `IS_DELAYED` = 1 if `Arr_Delay` > 15 minutes (FAA standard), and continuous regression target `Arr_Delay` in minutes.

Data leakage prevention: Post-flight breakdown columns (`Delay_Carrier`, `Delay_Weather`, etc.) are excluded as they are only populated after landing. `Dep_Delay` is included as it is observable at gate pushback, before the flight arrives.

4.2 Models

Random Forest (RF): An ensemble of 100 decision trees with maximum depth 12. Class imbalance is handled via `class_weight='balanced'`, which scales class weights inversely proportional to class frequencies.

Table 1: Feature set used for model training

Feature	Source	Description
`Dep_Delay`	Flights	Departure delay in minutes (known at pushback)
`Day_Of_Week`	Flights	Day of week (1=Monday, 7=Sunday)
`AIRLINE_ENC`	Flights	Label-encoded carrier code
`AIRPORT_ENC`	Flights	Label-encoded departure airport
`DEPTIME_ENC`	Flights	Time of day (morning/afternoon/evening/night)
`DIST_ENC`	Flights	Route type (short/medium/long haul)
`Flight_Duration`	Flights	Scheduled flight duration (minutes)
`W_TEMP`	Weather	Mean temperature at departure airport (°C)
`W_PRCP`	Weather	Precipitation (mm)
`W_WIND`	Weather	Wind speed (km/h)
`W_SNOW`	Weather	Snow depth (mm)
`W_PRES`	Weather	Atmospheric pressure (hPa)

Support Vector Machine / Regression (SVR): RBF kernel SVM for classification (SVC) and regression (SVR), with $C = 1.0$. Due to $O(n^2)$ scaling, SVR is trained on a 5,000-row subset per experiment.

Multi-Layer Perceptron (MLP): Two hidden layers (128 → 64 neurons), ReLU activation, Adam optimiser with learning rate 0.001. Early stopping with patience of 10 iterations on a 10% validation split prevents overfitting. Features are standardised with `StandardScaler` before MLP training.

4.3 Evaluation Protocol

Data is split 80/20 train/test with stratification on the classification target. All classification metrics are reported on the held-out test set. Regression metrics use the same test split.

5 Experiments & Results

5.1 Preliminary Results (D1)

Table 2 reports preliminary classification and regression performance on the 500k-row sample.

Table 2: D1 — Classification and regression results on 2023 US flight data

Model	Classification				Regression	
	Accuracy	Precision	Recall	AUC-ROC	RMSE (min)	R^2
Random Forest	0.912	0.801	0.798	0.950	14.39	0.955
MLP (128→64)	0.925	0.862	0.718	0.951	14.21	0.953

Both models achieve AUC ≈ 0.95 and $R^2 \approx 0.95$, improving on the benchmark of AlBassam & AlShahrani (2025) (AUC = 0.87) which did not incorporate weather features. The improvement of ≈ 0.08 AUC demonstrates measurable value from the weather dataset merge.

Random Forest has higher recall (0.798 vs 0.718), meaning it catches more actual delays. MLP has higher precision (0.862 vs 0.801), producing fewer false alarms. This reflects the classic precision-recall tradeoff between the two model families on imbalanced data, consistent with the findings of Grinsztajn et al. (2022).

5.2 Feature Importance

Random Forest feature importances show `Dep_Delay` dominating at $\approx 88\%$, consistent with Cai et al. (2022) who identify aircraft reuse as the primary delay propagation mechanism. Weather features collectively contribute $\approx 4-5\%$, with `W_WIND` and `W_TEMP` most informative, particularly for predicting extreme delay events.

6 Discussion

Why RF and MLP achieve near-identical AUC: The flight dataset is a heterogeneous tabular dataset with a mix of categorical and continuous features at different scales. Grinsztajn et al. (2022) predict exactly this outcome — on such data, tree ensembles match neural networks. The near-equal AUC validates this theoretical expectation empirically on real aviation data.

Departure delay as a feature: A reasonable concern is whether including `Dep_Delay` as a feature constitutes leakage. It does not — departure delay is observable at gate pushback, well before the flight arrives. This is the standard definition used in operational delay prediction systems. However, for a pre-departure prediction scenario (e.g., predicting delay before the flight departs), this feature would be unavailable. This experiment is planned for D2.

Limitations: (1) The 500k-row sample used in D1 may not fully represent the full-year seasonal distribution. (2) SVR is restricted to 5,000 rows due to quadratic scaling — full comparison at scale requires approximation methods. (3) Weather features are daily averages rather than hourly values, which may miss short-duration events like fog at departure time.

7 Conclusion

We present a flight delay prediction pipeline combining 2023 US flight operations data with daily NOAA weather observations. Random Forest and MLP both achieve AUC ≈ 0.95 and $R^2 \approx 0.95$ on both classification and regression tasks, outperforming the published benchmark of AUC = 0.87 that did not use weather features. The near-equal performance of RF and MLP is consistent with NeurIPS 2022 findings on tree models vs deep learning for tabular data. Future work (D2) will add SVR, apply coreset selection, pruning, and quantization to reduce model size and training data requirements while maintaining predictive performance.

References

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